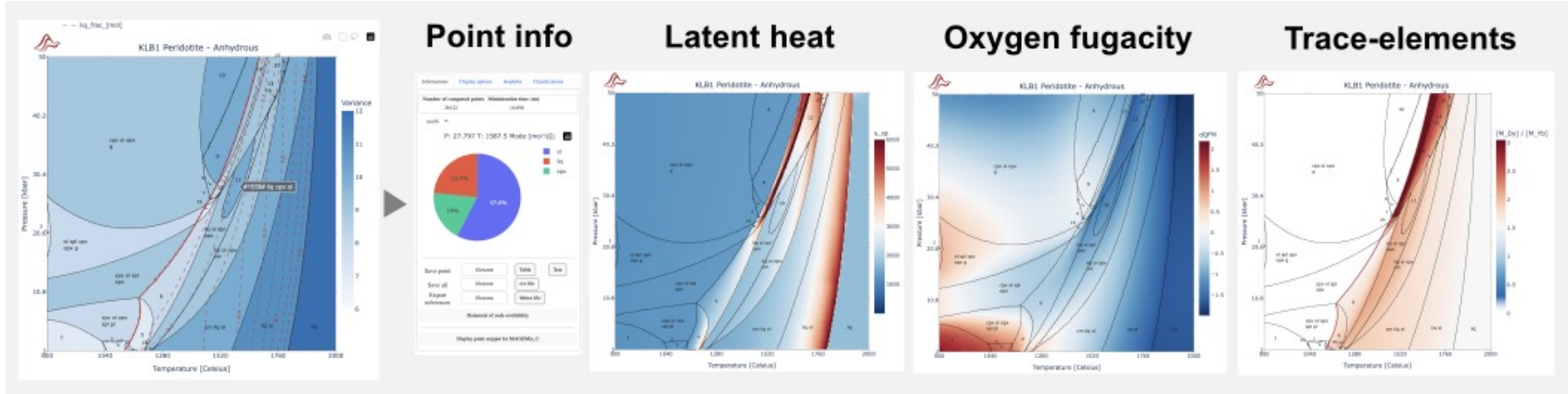
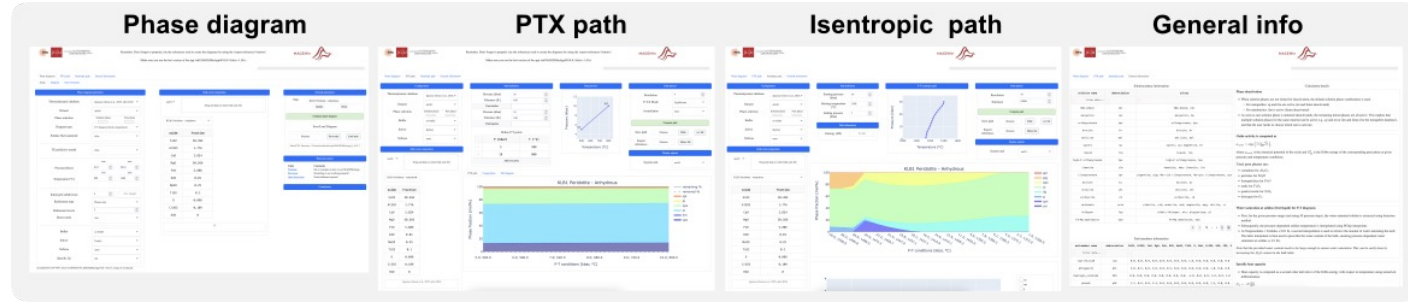


Phase Equilibrium Modelling with MAGEMin

Nicolas Riel - Johannes-Gutenberg University Mainz, Germany nriel@uni-mainz.de



Overview

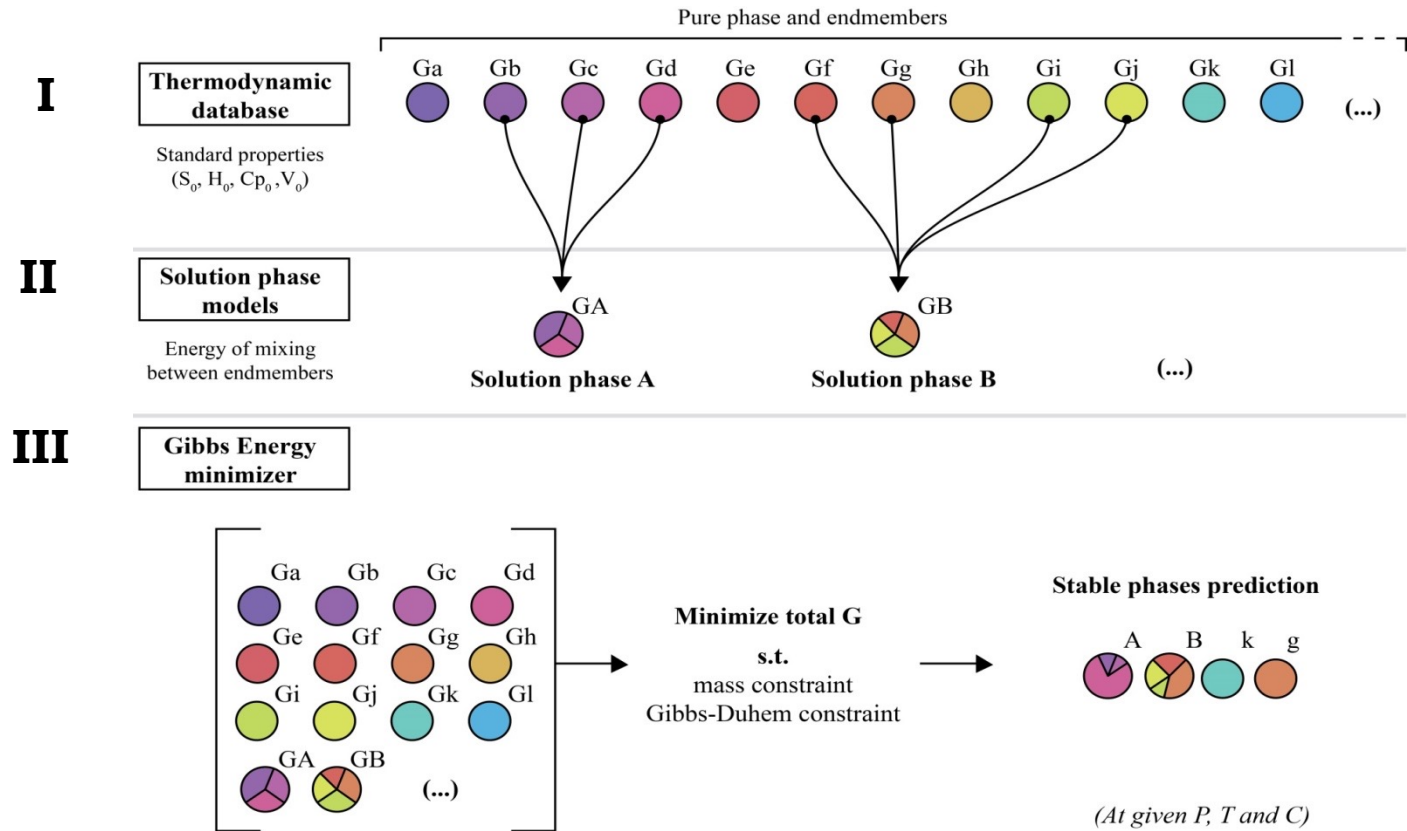


- General introduction and rationales
- Benchmarks and MAGEMin packages
- MAGEMinApp
- MAGEMin_C



https://computationalthermodynamics.github.io/MAGEMin_C.jl/dev/

How to compute a stable phase equilibrium?



e.g.,
 tc-ds6xx.txt for TC
 hp02ver.dat for Perple_X

Why a new Gibbs free energy minimizer?

Perple_X



MELTS

- Developed (in the 90's) to produce phase diagrams
- Natively not parallel
- Minimizations can often fail
- Not optimized for solution update
- Difficult to couple to geodynamic codes (written in Fortran)

MAGEMin: current stage and ongoing developments



Current stage

- MPI parallel C-library with several built-in database
(Holland et al., 2013; White et al., 2014; Green et al., 2016; Weller et al., 2024; Green et al., 2025; Evans & Forst, 2021)



- Low memory usage < 10 Mb
- Julia wrapper for petrological/geodynamic coupling (MAGEMin_C)
- Parallel web-browser Julia app (MAGEMinApp)
- Single core performance
~10 to 150 ms without initial guess (scales with #oxides and #phases)

Ongoing development

- Mantle dataset (Stixrude & Lithgow-Bertelloni, 2011)
- Algorithm upgrade to improve performances and stability (in review)
- Full Julia version
Better performances and flexibility (custom solution phase model, inversion framework for calibration etc.)

Application to magmatic system – wet system benchmark

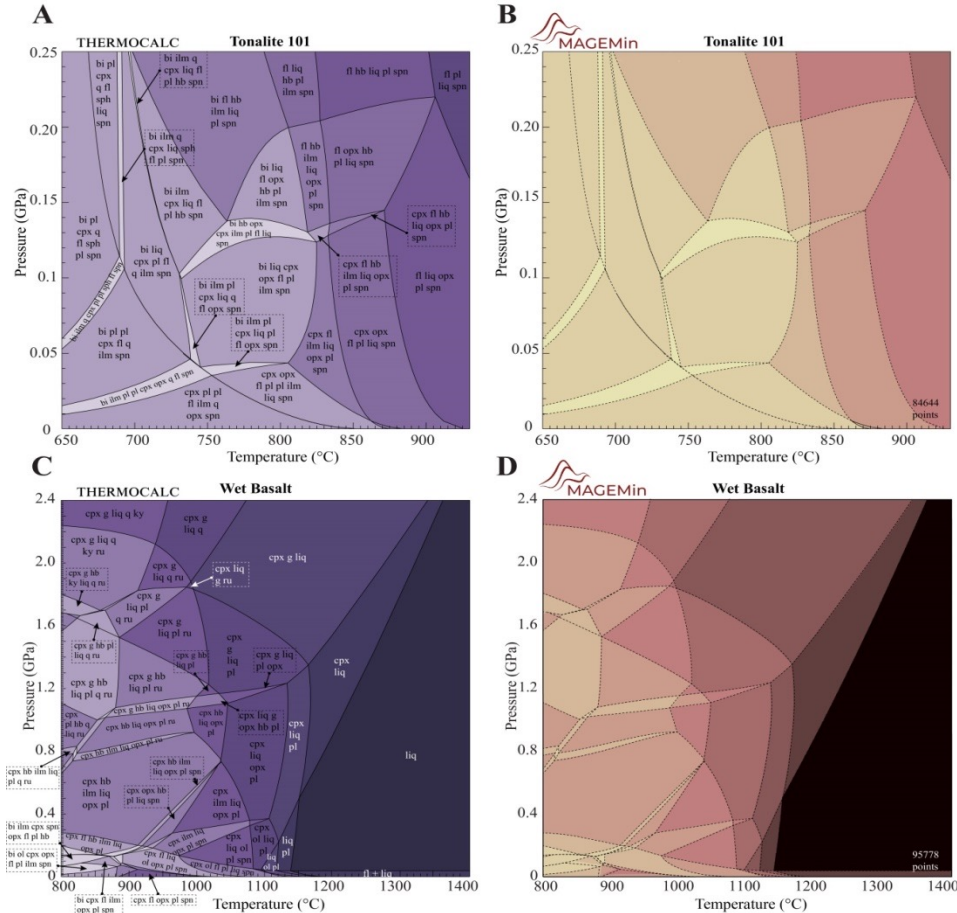
- NCKFMASHTCrO system, using the igneous database of Holland et al. (2018)

Tonalite

fluid oversaturated

Wet Basalt

fluid saturated



(Riel et al., 2022)

MAGEMin framework

(Mineral Assemblage Gibbs Energy Minimization)

MAGEMin



- MPI-parallel C code
- Point-wise minimization at given P-T-X
- Metapelite (White et al., 2014)
- Metabasite (Green et al., 2016)
- Igneous (Holland et al., 2018)
- Ultramafic (Evans & Forst, 2021)
- Mantle (Stixude & Lithgow-Bertelloni, 2010)

MAGEMin_C



- Julia wrapper of the C code
- Flexible programming interface
- Database calibration
- Geodynamic coupling

MAGEMinApp



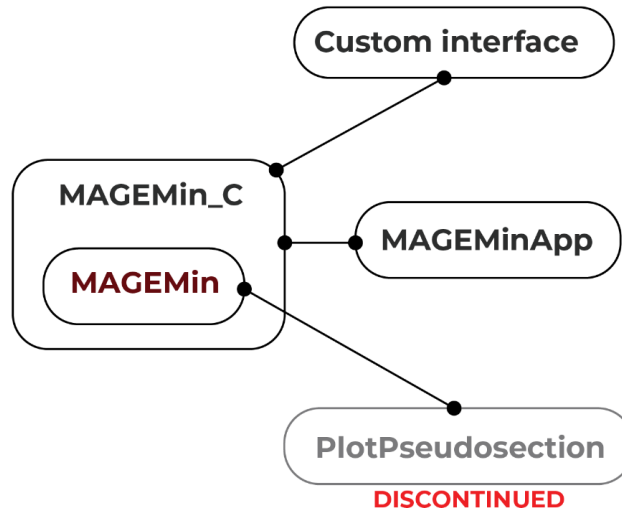
- Web browser app (graphic user interface)
- Parallel point-wise minimization
- PT, PX, TX and PT-X phase diagrams
- Auto labelling, contouring
- Fractional melting/crystallization paths

PlotPseudosection



- Matlab app (Graphic user interface)
- Parallel point-wise minimization
- PT, PX, TX diagrams
- Contouring
- PT paths
- Trace element partitioning for mafic to ultramafic systems

E.G., GEODYNAMIC COUPLING



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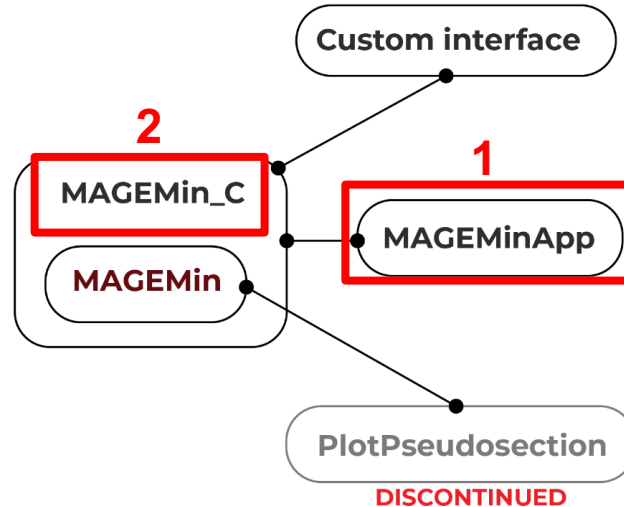
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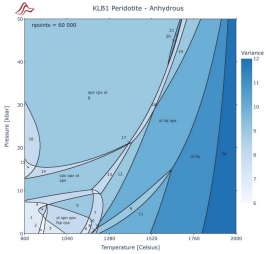


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E.G., GEODYNAMIC COUPLING



MAGEMin Github: ComputationalThermodynamics



nriel@uni-mainz.de



<https://github.com/ComputationalThermodynamics>
https://github.com/ComputationalThermodynamics/MAGEMin_C.jl
<https://github.com/ComputationalThermodynamics/MAGEMinApp.jl>
https://computationalthermodynamics.github.io/MAGEMin_C.jl/dev/

- Do not hesitate to contact me if anything is wrong/broken or needs to be added
- Everything is open source: you can contribute by creating pull-requests on Github!